

THE 2ND WORKSHOP & CHALLENGE ON SUBTLE VISUAL COMPUTING (SVC) @ CVPR 2026

# Text-guided Fine-Grained Video Anomaly Understanding

*Pixel Evidence to Language Reasoning*

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Video Anomaly Understanding

Subtle Visual Computing

LVLMS

Evidence-grounded QA



# Motivation

# Existing paradigms are fragmented

## Traditional IAD / VAD

Produces anomaly scores or heatmaps, but usually lacks semantic explanation.

Output: score / heatmap only

## General LVLMs

Can answer in language, but often fails to ground subtle cues at pixel level.

Output: judgment without explicit evidence

## LVLM + Diffusion hybrids

Combine text and visualization, but generated explanations may be unstable or inconsistent.

Output: judgment + generated visualization

## T-VAU

Couples pixel-level anomaly evidence with language reasoning.

Output: detection + localization + identification + trajectory + explanation

# The gap: anomaly understanding requires evidence

An anomaly should be  
**detected**, **localized** and **explained**.

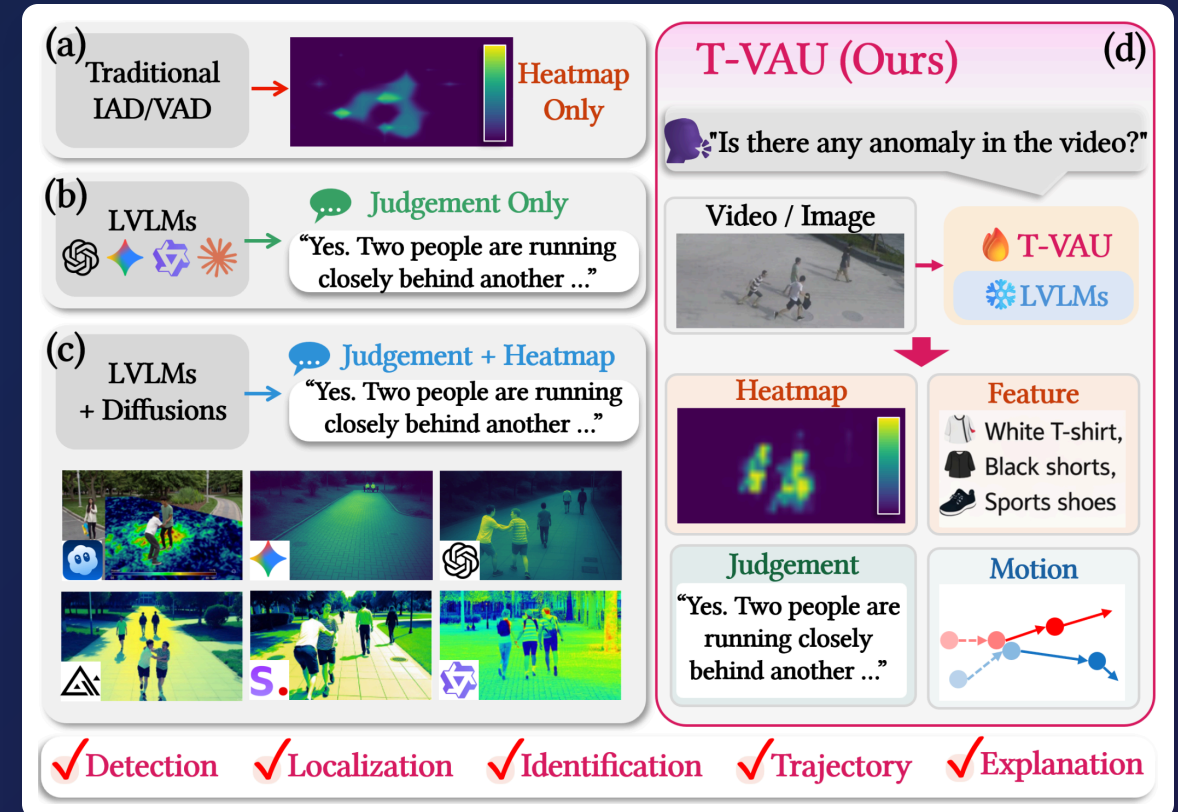
Fine-grained anomaly understanding asks the model to answer:

Is there any anomaly?

Where is the abnormal evidence?

What is the target's appearance?

How does the target move over time?





# What T-VAU adds

## 1. Pixel evidence

AHD produces threshold-free spatio-temporal heatmaps rather than only clip-level anomaly scores.

## 2. Region-aware prompting

RAE converts heatmap dynamics into local, global, and learnable prompt embeddings for the LVLM.

## 3. Target-level supervision

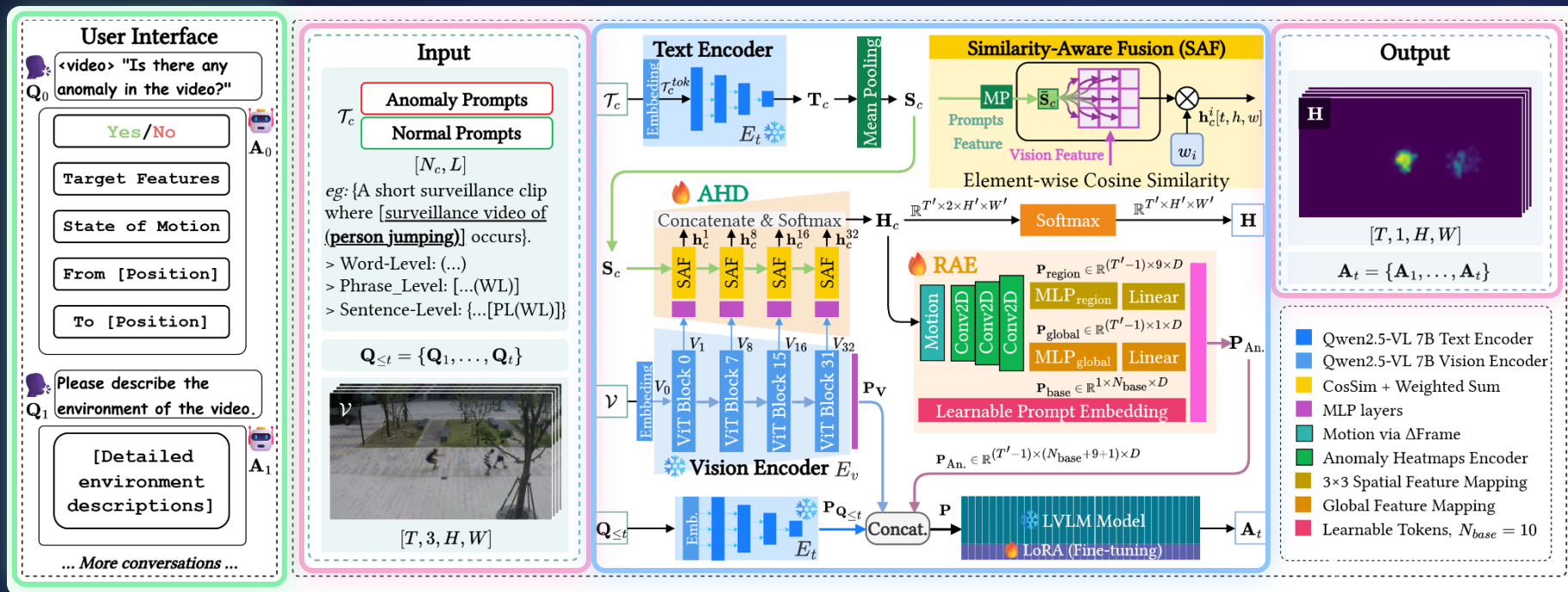
The dataset provides aligned annotations for appearance, localization, and motion trajectories.

*The core contribution is a closed loop: visual-text alignment → anomaly heatmaps → evidence-conditioned language reasoning.*

# Proposed Method

# T-VAU: closed-loop anomaly understanding

Given a video clip and multi-turn queries, T-VAU predicts:



Framework overview: Text Encoder + AHD + RAE + LVLm decoder.

**Pixel-level spatio-temporal heatmaps**

Explicit anomaly evidence,  
Anomaly Heatmaps

**Language responses**

Judgment, Appearance,  
Location, Motion,  
Trajectory

**Trainable lightweight modules**

Frozen Qwen2.5-VL 7B  
backbone + AHD + RAE  
+ LoRA prompt tuning.

# Key idea

## 1. Align

Use normal / abnormal text prompts to align intermediate visual tokens with semantic anomaly concepts.

Visual-text similarity  
(Element-wise cosine similarity)

## 2. Ground

Fuse similarity maps from multiscale ViT features to decode threshold-free heatmaps.

Anomaly Heatmap Decoder  
(Similarity-Aware Fusion + AHD)

## 3. Reason

Convert heatmap evidence into region-aware prompts and inject them into the LVLM.

Region-aware Anomaly Encoder  
(RAE + dialogue context)

*Low-level anomaly evidence becomes structured language reasoning.*

# Anomaly Heatmap Decoder (AHD)

## Purpose

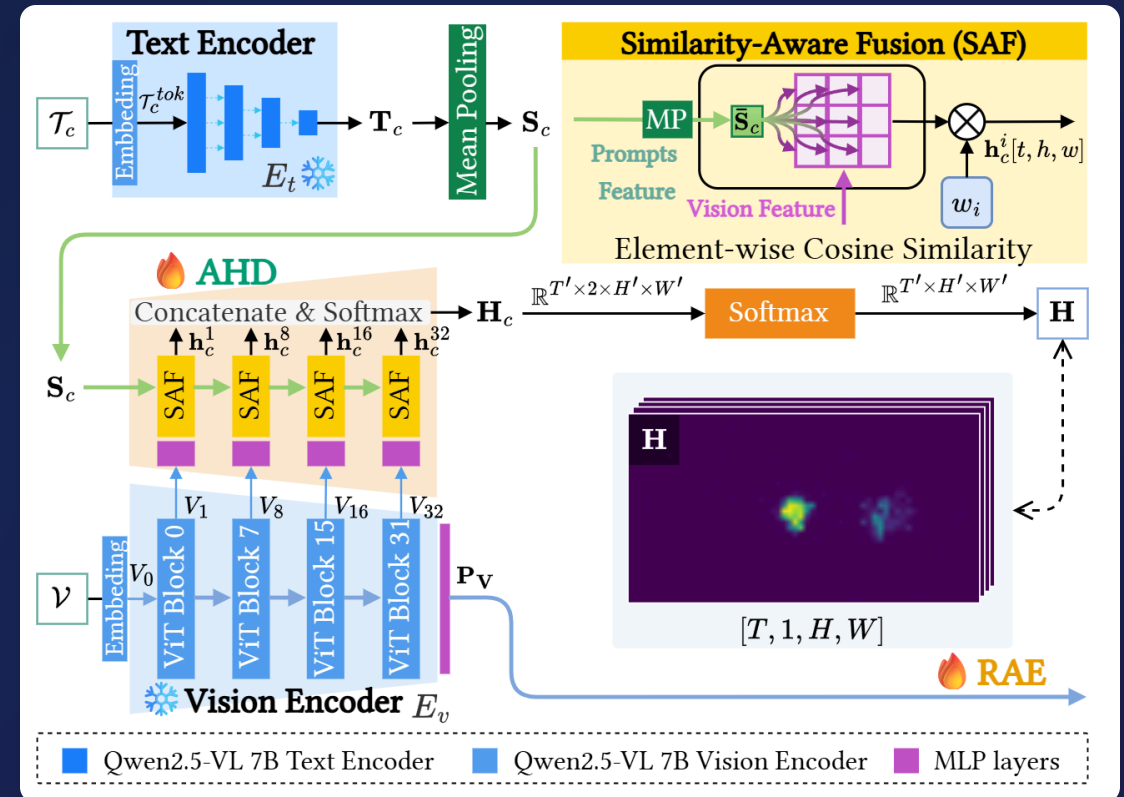
Generate pixel-level anomaly heatmaps by aligning visual tokens with text prompts.

Text prompts: normal vs. abnormal

Multiscale visual tokens: blocks 1, 8, 16, 32

Similarity-Aware Fusion:  
cosine similarity + weighted sum

Softmax anomaly channel  $\rightarrow$  heatmap  $\mathbf{H}$



Focus: Text Encoder + AHD + Similarity-Aware Fusion

# Region-aware Anomaly Encoder (RAE)

## Purpose

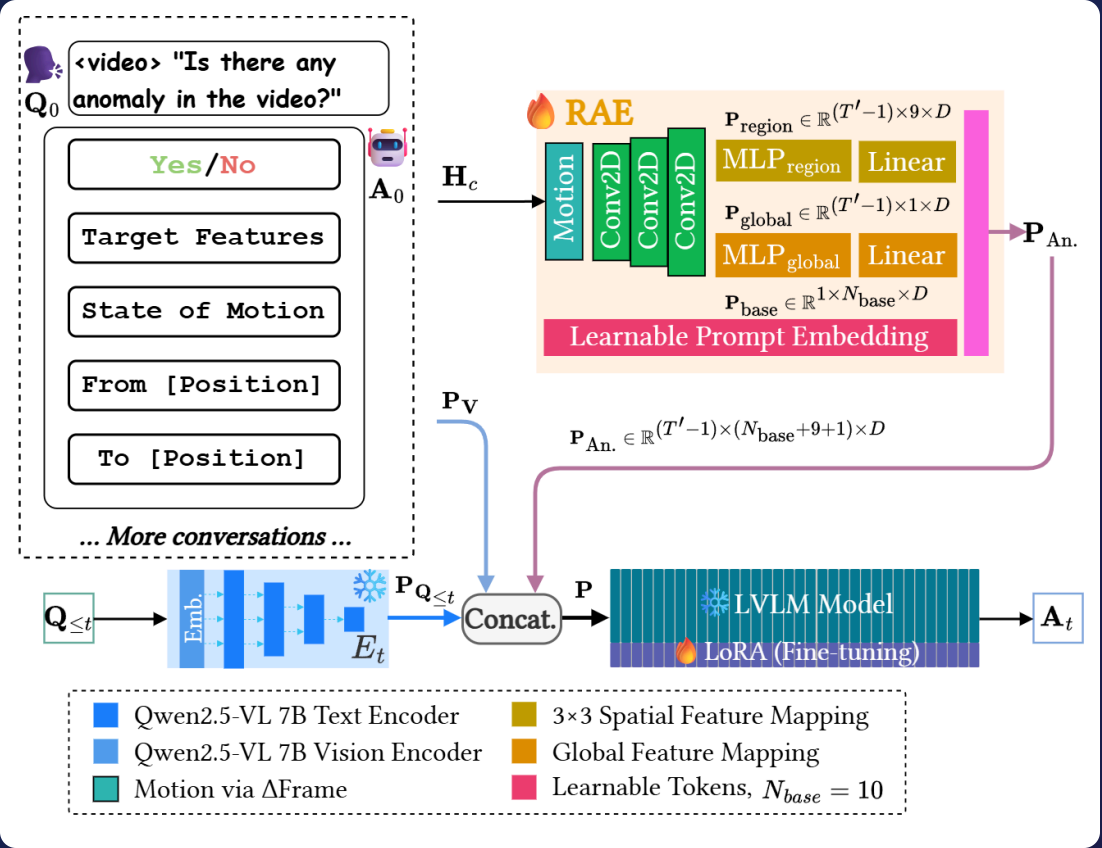
Transform heatmap evidence into prompts that the LVLM can use for multi-turn reasoning.

Temporal difference:  $\Delta H_c$  captures motion cues

Lightweight convolutional backbone extracts region-aware features

$3 \times 3$  regional pooling captures local evidence

Global pooling + learnable base tokens capture clip-level anomaly context



Focus: heatmap evidence → region-aware prompt embedding → LVLM response

# Multi-turn anomaly reasoning

## Dialogue flow

$Q_0$ : Is there any anomaly in the ?

$Q_1$ : Describe anomaly target features.

$Q_2$ : What is the motion state?

$Q_3$ : From where to where?



**Dialogue 1:**

Human: "Is there any anomaly in the video?"

AI: "Yes, there is **a person riding a bicycle** in the video."

Human: "Describe anomaly features in detail."

AI: "In the video, there is a lady riding a black bicycle and holding a white umbrella. Her physical features include: wearing a white top and black trousers, and carrying a pink backpack."

Human: "What is its movement trajectory?"

AI: "Her movement trajectory starts **from the right side of the picture**, **turns to the upper right corner** of the frame, and finally leaves the frame."

**Dialogue 2:**

Human: "Is there any anomaly in the video?"

AI: "Yes, in the video, there is **a silver van (SUV)** driving in from the left side of the frame."

Human: "Describe anomaly features in detail."

AI: "In the video, a silver van (SUV) suddenly **appears on the left side of the screen and quickly moves to the up**. The behavior of this vehicle is different from that of other objects and people because it suddenly appears and moves rapidly, while the surrounding crowd and environment do not show any obvious abnormal behavior or reaction."

*The language output is constrained by spatial and temporal evidence, reducing target drift and hallucinated explanations.*

# Fine-grained Dataset Construction



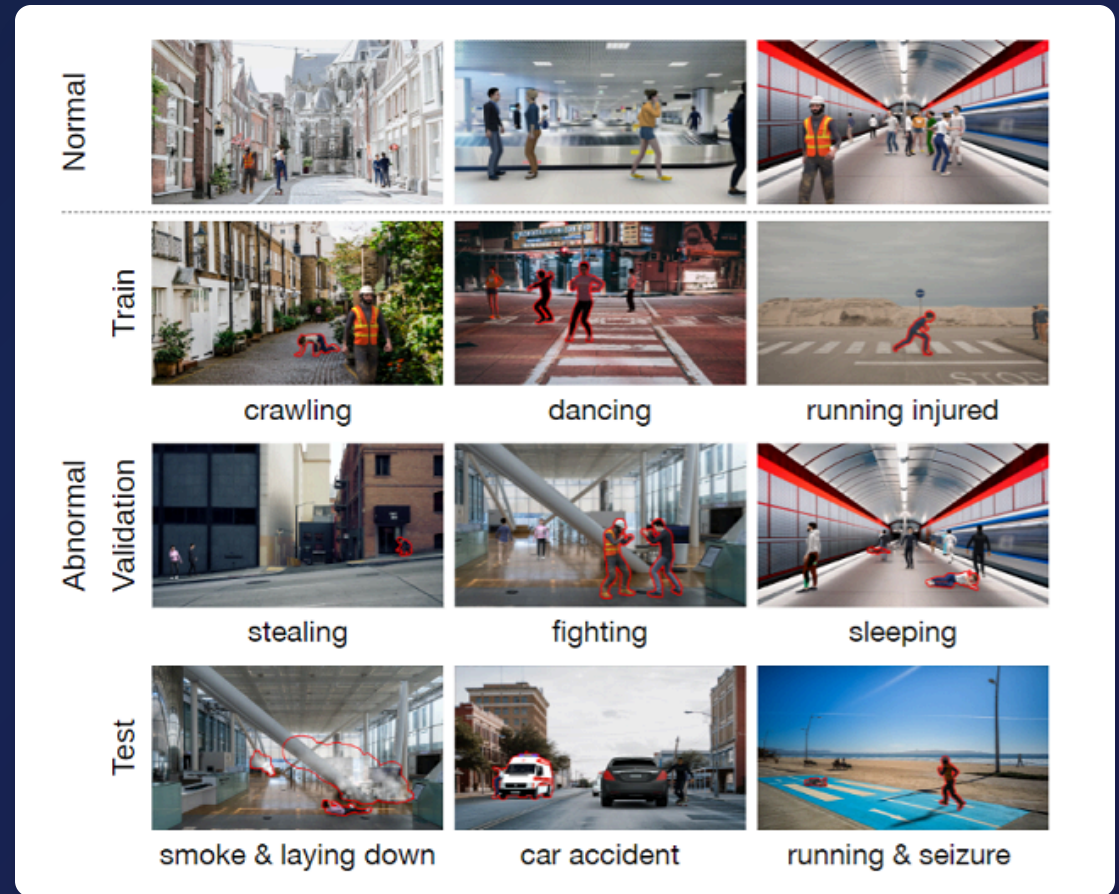
# Why construct a new supervision pipeline?

Pixel masks alone do not teach the model to explain.

VAD labels: anomaly score / mask

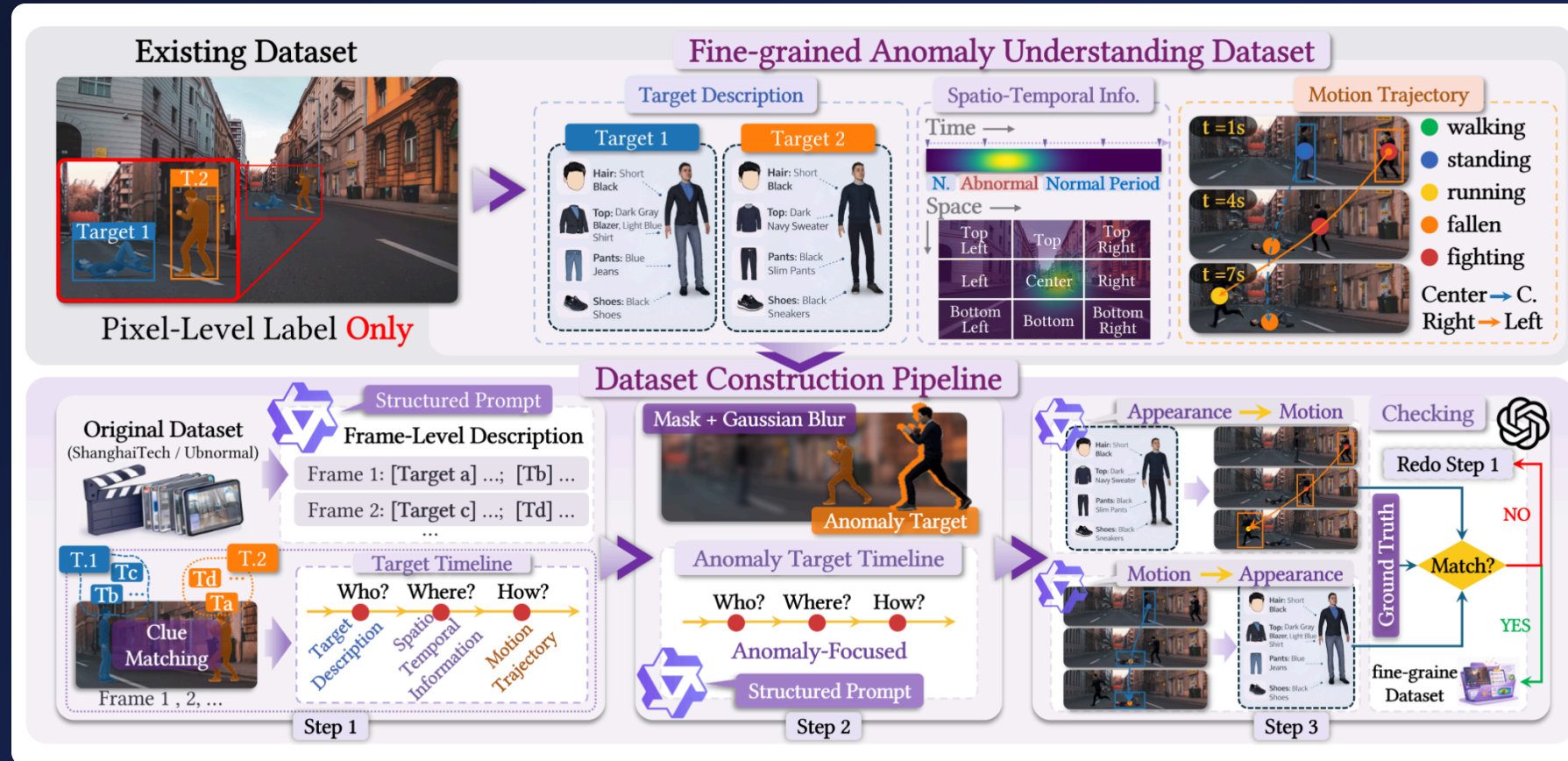
Needed by T-VAU: target identity, appearance, position, motion, trajectory

Solution: target-level video-text annotations derived from ShanghaiTech and UBnormal



UBnormal dataset examples (label mask only)

# Dataset construction pipeline



## Step 1

Frame-level structured prompting and temporal aggregation.

## Step 2

Mask-guided refinement with Gaussian-blurred background suppression.

## Step 3

Cross-modal consistency verification between appearance and motion.

# Resulting supervision

## Appearance

Target-level attributes such as clothing, object type, and visible distinguishing cues.

## Spatio-temporal localization

Frame-wise target location and abnormal period supervision.

## Motion trajectory

Direction, state changes, and target path over time.

ShanghaiTech: 4,108 train / 1,028 validation frame-wise annotations  
Target-aligned descriptions: 5,136 ShanghaiTech samples + 7,912 UBnormal samples

# Experiments

# Anomaly localization on UBnormal

## AHD achieves strong localization

Fine-tuned AHD improves micro-AUC, RBDC, and TBDC over prior baselines. Macro-AUC remains lower than the strongest multi-stream baseline.

94.8

Micro-AUC

87.8

Macro-AUC

67.8

RBDC

76.7

TBDC

| Method                         | Validation |         |        |        |
|--------------------------------|------------|---------|--------|--------|
|                                | AUC        |         | RBDC ↑ | TBDC ↑ |
|                                | Micro ↑    | Macro ↑ |        |        |
| Georgescu et al.               | 58.5       | 94.4    | 18.580 | 48.213 |
| Georgescu et al. (FT)          | 68.2       | 95.3    | 28.654 | 58.097 |
| Sultani et al. (PT)            | 61.1       | 89.4    | 0.001  | 0.012  |
| Sultani et al. (FT)            | 51.8       | 88.0    | 0.001  | 0.001  |
| Bertasius et al. (FT, SR=1/32) | 86.1       | 89.2    | 0.008  | 0.021  |
| Bertasius et al. (FT, SR=1/8)  | 83.4       | 90.6    | 0.009  | 0.023  |
| Bertasius et al. (FT, SR=1/4)  | 78.5       | 89.2    | 0.006  | 0.018  |
| AHD (1S)                       | 94.5       | 85.2    | 64.300 | 74.400 |
| AHD (FT)                       | 94.8       | 87.8    | 67.800 | 76.700 |

**Experimental results on UBnormal.** We report micro-/macro-averaged frame-level AUC, RBDC, and TBDC (%) for the baselines Georgescu et al., Sultani et al., and Bertasius et al. PT, FT, and 1S denote pre-trained, fine-tuned, and one-shot; SR denotes frame sampling rate. Although only Georgescu et al. supports anomaly localization, we report RBDC and TBDC for all baselines for completeness. Best results are highlighted in bold with a green background.



# Multi-turn dialogue evaluation

## RAE improves language grounding

One-shot evaluation compares representative LVLMs on target description, trajectory description, and Yes/No judgment.

ShanghaiTech: 62.67 Target BLEU-4, 88.84 Trajectory BLEU-4

ShanghaiTech Yes/No: 97.67%

UBnormal: 50.32 / 78.10 BLEU-4 and 89.73% Yes/No

| Method                   | Size ↓ | ShanghaiTech   |              |               | UBnormal       |              |               |
|--------------------------|--------|----------------|--------------|---------------|----------------|--------------|---------------|
|                          |        | BLEU-4 ± Std ↑ |              | Acc. ± Std ↑  | BLEU-4 ± Std ↑ |              | Acc. ± Std ↑  |
|                          |        | Target         | Trajectory   | Yes/No        | Target         | Trajectory   | Yes/No        |
| Qwen2.5-VL (zero-shot)   | 7B     | 18.74 ± 0.82   | 27.33 ± 1.05 | 61.03 ± 0.38% | 16.20 ± 0.85   | 24.18 ± 1.08 | 65.62 ± 0.41% |
| Qwen2.5-VL (one-shot)    | 7B     | 50.42 ± 0.58   | 78.91 ± 0.72 | 92.36 ± 0.24% | 44.35 ± 0.62   | 70.82 ± 0.75 | 87.24 ± 0.27% |
| LLaVA-1.6 (one-shot)     | 7B     | 47.68 ± 0.60   | 75.42 ± 0.74 | 91.07 ± 0.26% | 42.11 ± 0.64   | 68.07 ± 0.78 | 85.91 ± 0.28% |
| MiniCPM-V 2.6 (one-shot) | 7B     | 52.34 ± 0.54   | 80.41 ± 0.69 | 93.11 ± 0.23% | 46.70 ± 0.59   | 72.88 ± 0.73 | 86.94 ± 0.26% |
| Idetics2 (one-shot)      | 8B     | 44.29 ± 0.62   | 73.84 ± 0.76 | 90.12 ± 0.27% | 39.51 ± 0.66   | 65.92 ± 0.80 | 84.03 ± 0.29% |
| InternVL (one-shot)      | 8B     | 55.73 ± 0.50   | 82.65 ± 0.66 | 94.28 ± 0.22% | 49.84 ± 0.55   | 71.63 ± 0.70 | 88.65 ± 0.24% |
| RAE (Ours) (one-shot)    | 7B     | 62.67 ± 0.45   | 88.84 ± 0.53 | 97.67 ± 0.12% | 50.32 ± 0.49   | 78.10 ± 0.58 | 89.73 ± 0.18% |

One-shot evaluation results of representative LVLMs on our constructed dataset. “Size” denotes the number of model parameters in billions. BLEU-4 is reported for Target and Trajectory, and Accuracy for Yes/No. All metrics are in percentages. Our RAE performs best across tasks.

# Ablation: both modules are necessary

## Complementary roles

**AHD** supplies pixel-level evidence.

**RAE** makes this evidence usable by the LVLM.

Removing **AHD** makes heatmap metrics inapplicable.  
Removing **RAE** leaves heatmaps but removes evidence-conditioned dialogue.

| Method              | Size ↓          | Heatmap (UBnormal) ↑ |                    | BLEU-4 (ShanghaiTech) ↑ |                     | Acc. (S.T.) ↑        |
|---------------------|-----------------|----------------------|--------------------|-------------------------|---------------------|----------------------|
|                     | Parameters      | RBDC ± Std           | TBDC ± Std         | Target ± Std            | Trajectory ± Std    | Yes/No ± Std         |
| T-VAU w/o AHD       | 8299.71M        | --                   | --                 | 61.82 ± 0.42            | 85.47 ± 0.51        | 95.38 ± 0.18%        |
| T-VAU w/o RAE       | 8317.13M        | 67.8 ± 0.36          | 76.7 ± 0.41        | --                      | --                  | --                   |
| T-VAU w/o AHD & RAE | 8274.74M        | --                   | --                 | 61.82 ± 0.42            | 85.47 ± 0.51        | 95.38 ± 0.18%        |
| <b>T-VAU</b>        | <b>8324.67M</b> | <b>67.8 ± 0.36</b>   | <b>76.7 ± 0.41</b> | <b>62.67 ± 0.45</b>     | <b>88.84 ± 0.53</b> | <b>97.67 ± 0.12%</b> |

**Table:** Results of different **T-VAU** variants. "Parameters/Size" denotes the number of model parameters in millions. Heatmap metrics include RBDC and TBDC. BLEU-4 is reported for both Target and Trajectory. Accuracy is evaluated on Yes/No classification.

# Qualitative Results



# Evidence-grounded multi-turn QA



"Is there any anomaly in the video?"

"Yes, there is **a person riding a bicycle** in the video."



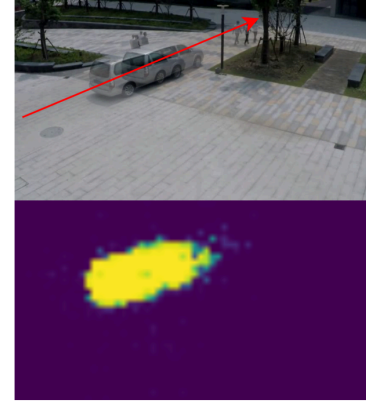
"Describe anomaly features in detail."

"In the video, there is a lady riding a black bicycle and holding a white umbrella. Her physical features include: wearing a white top and black trousers, and carrying a pink backpack."



"What is its movement trajectory?"

"Her movement trajectory starts **from the right side of the picture, turns to the upper right corner** of the frame, and finally leaves the frame."



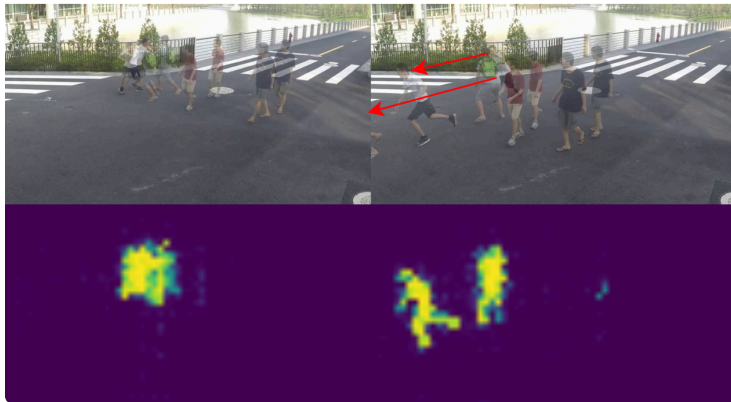
"Is there any anomaly in the video?"

"Yes, in the video, there is **a silver van (SUV)** driving in from the left side of the frame."



"Describe anomaly features in detail."

"In the video, a silver van (SUV) suddenly **appears on the left side of the screen and quickly moves to the up**. The behavior of this vehicle is different from that of other objects and people because it suddenly appears and moves rapidly, while the surrounding crowd and environment do not show any obvious abnormal behavior or reaction."



"Is there any anomaly in the video?"

"Yes. In the video, there is indeed an abnormal situation. Specifically, in frames 8 and 9, there is **a person wearing a white top and grey shorts, carrying a green schoolbag, walking with other pedestrians**. However, in frame 10, they suddenly start running, and their position is significantly different from the trajectories of the other pedestrians. This can be regarded as abnormal behavior."



"Describe anomaly features in detail."

"The abnormal characteristics are as follows:  
- Wear a white top and grey shorts.  
- Carrying a green schoolbag.  
- Walk with other pedestrians between frames 8 and 9.  
- Suddenly start running in frame 10.  
- The position is inconsistent with the trajectories of other pedestrians in frame 9."

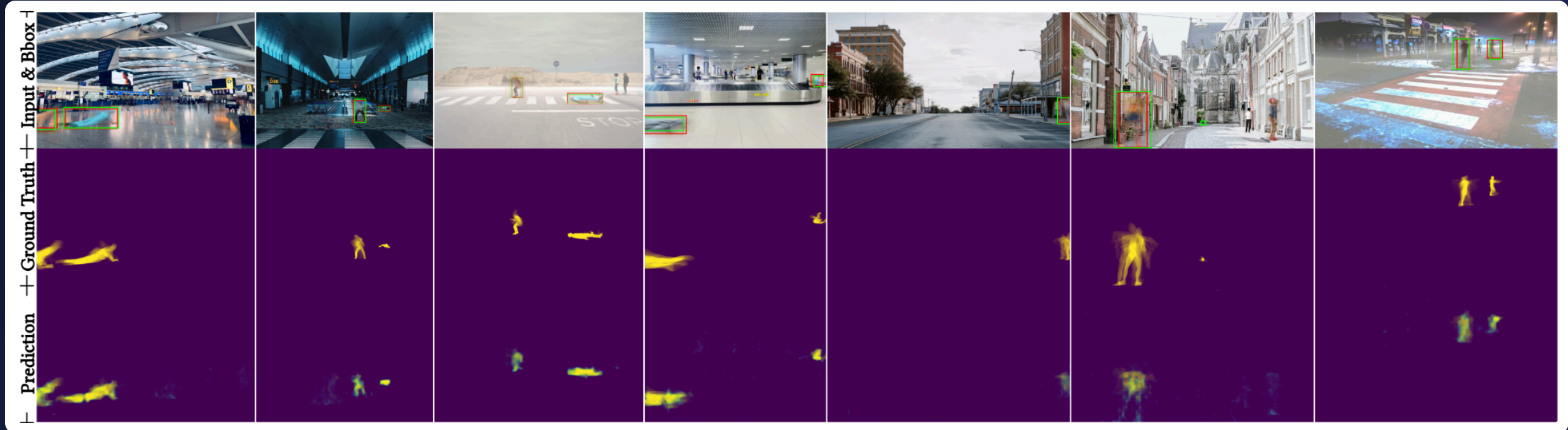


**AHD** localizes the abnormal target rather than the whole scene.

**RAE** guides target-specific appearance and motion descriptions.

The dialogue remains consistent across judgment, features, and trajectory.

# Trajectory consistency



*Accumulated predictions follow the ground-truth anomaly regions over time, supporting motion and trajectory reasoning.*

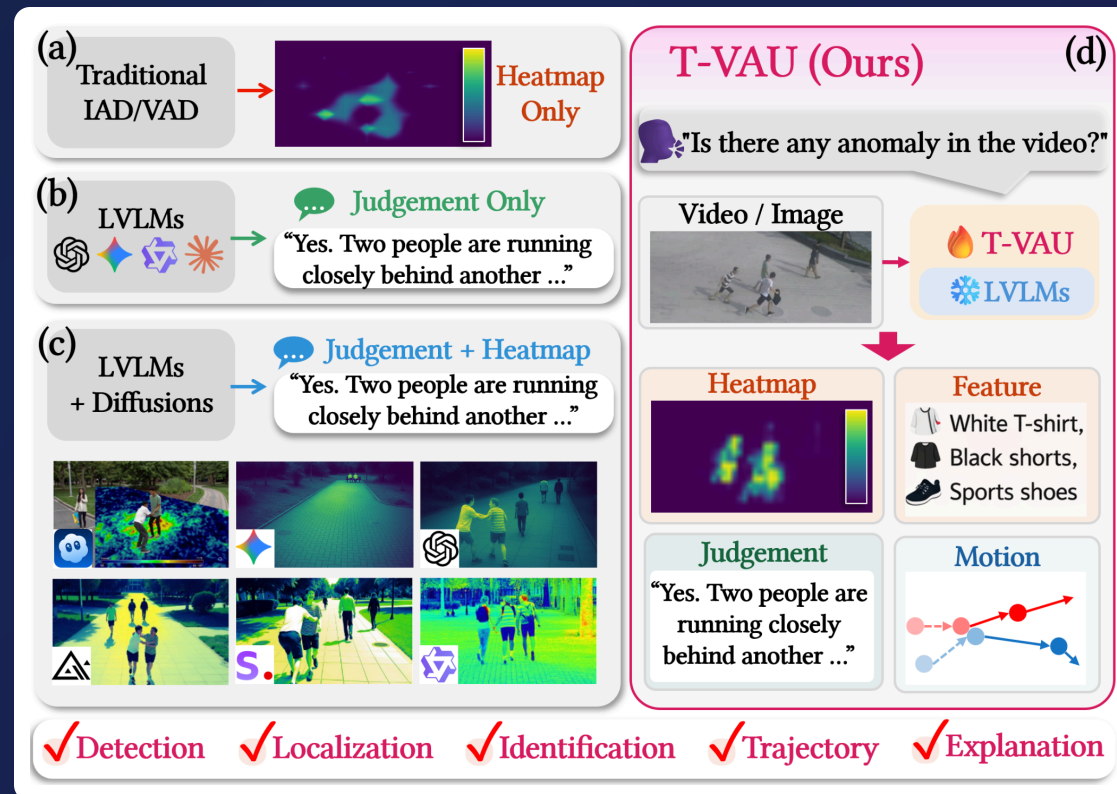
# Takeaway

T-VAU closes the loop from **pixel evidence** to **language reasoning**.

**AHD**: threshold-free spatio-temporal anomaly heatmaps

**RAE**: region-aware and motion-aware prompt injection

**Dataset**: target-level appearance, localization, and trajectory supervision



# Thank you

